Project Report: Machine Learning Techniques and Architectures on a Noisy SuperCDMS Detector Dataset

Zachary Allen Robert St. Clair Utsab Mitra
Georgia Institute of Technology
zallen34@gatech.edu, rclair7@gatech.edu, umitra7@gatech.edu

Abstract

Accurately reconstructing the interaction locations of Weakly Interacting Massive Particles (WIMPs) which are candidate dark matter particles in SuperCDMS germanium detectors is essential for separating signal events from background noise. This task is complicated by noisy pulse signals and limited labeled data. Prior benchmark work by the FAIR research group at the University of Minnesota establishes a performance baseline using simple deep neural networks with two-hidden layers. achieving a held-out subset (HOS) RMSE of 1.741 mm on 7,151 interactions across 13 different locations on the detector along a radial path. We investigate more sophisticated architectures and training strategies to improve on this baseline, including a convolutional neural network and an attention-based sequential model. We augment data by adding noise profiles to expand the dataset approximately three-fold and evaluate its per-class RMSE in a series of experiments with simpler models. The CNN architecture is an implementation of a two-branch model that separates per-detector pulse shape from spatial arrival patterns across detectors into dedicated processing streams with a shared convolutional encoder across five detector channels. We implement a multi-head attention encoder model with modified softmax functions that learn thermodynamic inspired parameters.

The CNN yields a HOS RMSE of 1.611 mm, a 7.4% improvement over the benchmark, while the thermal attention model ties the HOS error and reduces the RMSE to 0.4621 mm when a 80/20% holdout split is used. Across all methods, our findings suggest that architectural choices can be more effective than scaling model depth.

1. Introduction

The Super Cryogenic Dark Matter Search (SuperCDMS) experiment searches for dark matter candidates by detecting vibrational signals produced when candidate dark matter particles interact with a silicon and germanium particle detector at approximately 30 mK. Five regions of the sensor's surface output the impulse response of an incident interacting particle. Accurately reconstructing the 2D positions of these interactions is essential to distinguish potential signals from background noise.

In [1], the FAIR research group at the University of Minnesota employed basic machine learning techniques on a dataset of 7,151 labeled interactions across 13 known radial positions, using 19 hand-engineered pulse shape features as model inputs. Among the various architecture investigated, a two-hidden-layer fully connected neural network achieved the best generalization on a held-out subset of positions with a root mean squared error (RMSE) of 1.741 mm, while deeper architectures degraded this result - for example, a ten-hidden-layer produced 7.072 mm RMSE - indicating that network scaling causes overfitting. This benchmark concluded that novel architecture or feature engineering were needed to improve the performance.

We seek to improve upon this work by employing a number of deep learning methods and comparing the effectiveness of tuned learning models on the predictions of these interaction locations on the surface of the sensor. We employ the following strategies: (a) Class-based RMSE Evaluation (b) A two-branch convolutional neural network with a shared pulse encoder that processes spatial and temporal streams separately before joining. (c) A self-attention based model with learnable softmax functions derived from statistical physics in each attention head, with an encoder-only transformer.

Method (a) achieved a best RMSE of 1.1622 mm, (b) achieved a best RMSE of 1.611 mm, and (c) achieved a best RMSE of 0.4621 mm.

Producing an error below the 1.741 mm RMSE precision benchmark is an important step towards integrating machine learning methods into the hunt for dark matter candidates and the improvement of the fidelity of existing detectors. In this paper, we build upon the existing FAIR work and showcase notable reduction of error on the existing dataset.

2. Dataset

The work in [1] provides us with two datasets describing 7,151 interaction events: a full dataset and a reduced dataset. The full dataset contains pulse amplitudes and 16 chronological amplitude-threshold crossing times recorded across five sensor channels spanning the rise and fall of each sensor's pulse, while the reduced dataset contains 19 calculated pulse shape features as inputs. Both datasets contain particularly small quantities, such as signal threshold times and pulse width measurements, most of which are on the order of 10^{-6}. The work in [1] only utilizes the reduced dataset, however with a dataset on the smaller side with only 13 interaction positions along the sensor radius, we believe that using the reduced dataset could result in the loss of useful data present in the full pulse data. Some of the models we present use both datasets and we report the more effective of the two.

While conducting our work, we identified an issue with the full dataset: the labels associated with each column were clearly jumbled. The amplitude data was identified by hand, while the 5 sets of 16 time-ordered columns required the use of a stochastic optimizer to sort. A simple optimizer using an energy objective function was coded to swap columns between five groups until all columns were properly ordered. The sorted data was then used to recalculate the reduced data set to verify that the unscrambling had been done properly. The code for this process is provided in the supplementary files that accompany this paper.

2.1. Data Augmentation

The limited number of data entries can be a limiter of the effectiveness in training our models. We attempt to remedy this by augmenting the data to add more reasonable pseudo-measurements. This was done by adding noise and then re-sampling the signal. Given the nature of the experiment and the location of the detector being far underground, we do not expect the noise to have a large high-frequency contribution due to attenuation of signals through the ground. We added amounts of Brownian (Brown), Flicker (Pink) noise, and Shot noise to the signal. The magnitude of the noise was varied to generate augmented pulses with varying signal to noise ratios caused by a tunable mixture of these noise contributions. Our idea was to help the model become more robust and more catered to elusive particles if the signal to noise ratio of the data points is decreased.

Each pulse in the original dataset was interpolated using a piecewise cubic Hermite interpolating polynomial (PCHIP) to ensure the data was not overshot by interpolation methods. This interpolator defined in [2] is continuous and first-order differentiable which is perfect for the pulse data. Noise is then generated and added to the interpolated curve, and threshold sampling is conducted at the same pulse rise and fall amplitude thresholds as the original dataset. To get the reduced dataset, we calculate the same pulse shape parameters as the original work from [1]. A visual depiction of the augmentation is shown in Figure 1.

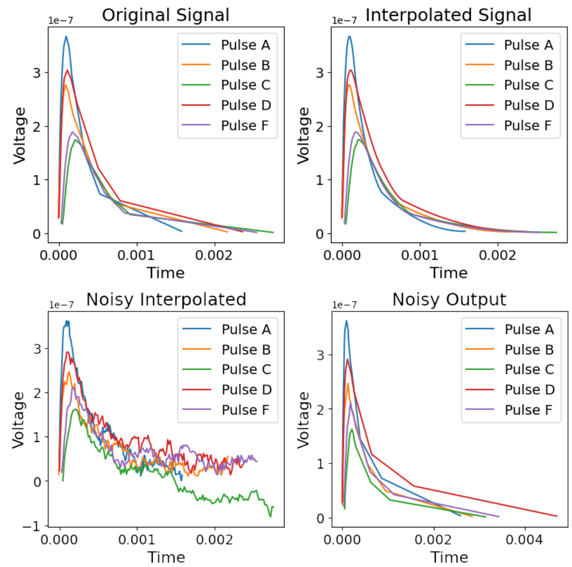

Figure 1. The data augmentation process shown in successive steps. We start from the original pulse signal, create an interpolated curve, add noise and then re-sample the curves.

3. Approach

In this section, we detail three distinct methodologies developed to improve upon the FAIR benchmark. First, we establish a Class-based RMSE Evaluation framework to analyze how model complexity impacts extrapolation at specific detector locations. Second, we introduce Thermal Attention, a transformer-based approach that utilizes a custom softmax function derived from statistical physics to process temporal pulse data. Finally, we present a Two-Branch Convolutional Neural Network designed to separately encode intra-channel pulse shapes and cross-channel spatial arrival patterns.

3.1. Class-based RMSE Evaluation

While the original results of the FAIR research group evaluated model performance using overall test set RMSE, we hypothesized that this aggregate metric may obscure important performance characteristics, particularly regarding extrapolation capability. Inspection of the held-out set (HOS) reveals an interaction location of $y = -41.900$ mm that falls outside the range of the training data (spanning 0.0 to -41.010 mm). This presents a subtle but important extrapolation challenge distinct from the interpolation required for the other two held-out locations at $y = -12.502$ mm and $y = -29.500$ mm. To gain deeper insight into model behavior, we decomposed the overall test RMSE into per-class RMSE for each of the three held-out impact locations.

We fit four model types (Linear/Ridge/Lasso/DNN) across three dataset configurations: reduced (19 engineered features), augmented (19 features, 3x samples via synthetic generation), and full (80 raw timing markers). Within each configuration, we varied amplitude feature inclusion (5 additional features), PCA dimensionality reduction, and DNN depth (2/5/10 layers), yielding 34 total experiments. All models were evaluated using the same train/validation/test split as the original paper to ensure direct comparability. These experiments asses whether models with similar overall RMSE exhibit different spatial error patterns across individual locations, and how feature engineering and model complexity affect extrapolation capability.

3.2. Convolutional Neural Network

For the implementation of the Convolutional Neural Network (CNN), we focused on three targeted interventions: a physically meaningful feature engineering, an architecture tailored to the input semantics, and a training strategy addressing the discrete nature of the training data. We started with the full temporal dataset, as described in Section 2.1, and took the 80 raw timing markers across 5 detectors [N, 80]. It was transformed into a structured [N, 5, 16] tensor. This representation explicitly captures physically meaningful relationships while removing irrelevant variations.

- **Intra-Channel Deltas (Rows 0-4):** The first entry of each detector row represents that detector's *r10* arrival time referenced to Channel A's *r10*. The remaining 15 entries are consecutive differences between adjacent markers (e.g. *r20* - *r10*), isolating the pulse shape.
- **Spatial Arrival Encoding (Row 5):** A 5-element vector contain each detector's *r10* minus Channel A's *r10*, zero-padded to 16 entries. This captures relative phonon propagation delays across the detector, constituting the primary spatial localization signal.

Standard CNNs treat input channels symmetrically, which is inappropriate given our structurally distinct data types. We implemented a Two-Branch architecture [5, 6] that processes pulse shapes and spatial patterns through dedicated units:

- **Pulse Branch:** Uses a shared 1D convolution encoder (32 filters, kernel size 5) with batch normalization and ReLU activation. An AdaptivePool1d collapses the time axis to two points, forcing the model to learn summary statistics of the pulse shape. We employ a weight-sharing scheme [4] where the encoder is applied identically to all five detectors, training a single pulse representation on five times the available data to improve generalization.
- **Spatial Branch:** Processes the 5-element cross-channel data through a two-layer MLP with 16 hidden units.
- **Fusion and Regression:** The resulting 80-dimensional pulse embedding and 16-dimensional spatial embedding are concatenated and passed through a regression head (linear layers, ReLU, and 0.2 dropout) to produce the scalar position prediction. The model comprises only 4,769 parameters to prevent overfitting.

3.3. Thermal Attention

To construct our attention model, we begin with the architecture laid out in [7]. Because we seek to map the inputs to a singular output, we strip the entire decoder section of the transformer out while still keeping the self-attending encoder. We also deviate from the model by utilizing unique heads that learn their own attention models.

The SuperCDMS device utilizes crystal lattices to detect incident dark matter candidate particles. The disturbance of a crystal lattice can be naively modeled using statistical mechanics, by defining such a disturbance as an imaginary vibrational particle called a phonon. [3] shows that a simplified version of the phonon model can be modeled using a Maxwell-Boltzmann probability distribution:

$$P_i = \frac{e^{g_i \beta (E_i - \mu)}}{Z} \tag{1}$$

Where i is a specific phonon mode and P_i is the probability that the phonon will have the corresponding energy E_i. The degeneracy, or number of particles in the specific phonon mode is g_i. β and μ are thermodynamic constants that map to temperature and potential energy, and Z is the following partition function:

$$Z = \sum_{n=0}^{\infty} e^{\beta (E_n - \mu)} \tag{2}$$

Note that in this form, P_n is exactly a softmax function when g_i and β are set to one and μ is set to zero. Because of this, we utilize an attention model that incorporates the Maxwell-Boltzmann distribution as a custom softmax attention with learnable parameters β and μ. Since the data only has one impulse per sensor, we set g_i equal to one.

This leaves the final "thermal" softmax equation as

$$P_i = \frac{e^{\beta (E_i - \mu)}}{\sum_{n=0}^{N} e^{\beta (E_n - \mu)}} \tag{3}$$

We insert Equation 3 into the standard architecture for an attention head, and set up the model so that each head learns its own β and μ. The full input dataset is divided into 5 channels (dimensions) based on the time sequence sensor measurements and fed into the model. Due to their small magnitude, the times are scaled to a domain of [0,1) to prevent small gradients. Each dimension receives positional embeddings, since the temporal position of each entry is important information about how the original interaction affects each sensor area. Each embedded channel is passed to an attention head and then combined. The combined vector is sent through a two-layer feed forward network followed by a normalization layer and then a final multi-layer perceptron to map the the transformer output to a single value. We also incorporate a residual connection to facilitate gradient flow.

The output is then passed through a sigmoid function and multiplied by -50. This enforces that the model say in the range of the dataset (0 to -41.9). As opposed to a standard attention model, this model allows for each attention head to learn the shape of its softmax, allowing the model to widen or narrow this attention distribution as it learns.

Since the thermal attention encoder network takes advantage of time-ordered sequential data, we utilized only the full dataset for this model. The model was provided two training sets of data: one with the original HOS and one with a 80/20 percent split off training and holdout data. The training set was further split into train and validation sets, regardless of the HOS.

We tuned three of the four tunable parameters for this model: the base learning rate, the thermal parameter learning rate, and the hidden dimension. We opted to keep the number of heads for this attention model fixed at 5 so that each input channel had only one attention head that learned its own thermal parameters for that input channel. The model was trained over 100 epochs and RMSE was used as an evaluation criterion.

4. Experiments and Results

4.1. Class-based RMSE Evaluation

A key finding of the class-based RMSE evaluation is that extrapolation difficulty varies by feature richness. At $y = -41.900$ mm, RMSE ranged from 0.93 mm (Lasso, full data) to 5.15 mm (Linear, full data), demonstrating that comprehensive timing information enables accurate extrapolation when properly regularized. In contrast, the reduced dataset achieved 1.15-1.95 mm and the augmented data further degraded to 1.92-2.03 mm, suggesting synthetic augmentation introduces noise rather than meaningful diversity.

Additionally, introducing PCA dimensionality reduction on the three datasets by converting the raw features into a reduced set of principal components retaining at least 90% of the observed variances increased per-class RMSE by 58-150% across all locations. This demonstrates that variance preservation does not guarantee task-relevance and PCA eliminates correlation patterns between channel pairs that encode positional information. Similarly, including the 5 amplitude features for each channel also degraded the best achievable extrapolation at $y = -41.900$ mm from 0.93 mm RMSE (Lasso, full data) to 1.24 mm (Ridge, full data + amplitude), confirming the benchmark finding that calibration drift introduces systematic errors outweighing information gains.

Similar to the results of the original paper, the best DNNs were with 2-layers with the overall best achieving 1.58 mm overall RMSE using the reduced dataset. This is competitive with Lasso (1.52 mm), but inferior at extrapolation (1.75 mm vs 1.15 mm for Ridge). The 2-layer DNN on full data achieved 2.63 mm overall RMSE, showing no benefit from additional features.

In summary, there are noticeable differences in the per-class test RMSE values that should be considered when evaluating models for this task, as the exterior location $y = -41.900$ mm is outside the range of the training data and can disproportionately influence the overall test RMSE. Physics-informed timing parameters achieve competitive performance compared to raw makers, while the creation of an augmented data set to increase sample size provides no measurable benefit. The extrapolation RMSE for the best configuration of each model based on overall test RMSE is shown in Table 1.

4.2. Convolutional Neural Network

Following the original benchmark, we reserved the positions $y =$ [-12.502 mm, -29.500 mm, -41.900 mm] for the Held-Out Subset (HOS). Because the training data consists of only nine discrete y-values, models often learn a "soft classifier" that fails to interpolate accurately between large gaps (e.g., the 12 mm gap surrounding the -29.500 mm HOS position). To address this, we introduced intermediate y-values using a sampling scheme inspired by $Beta(\alpha, \alpha)$ distributions [8] to provide inter-position supervision and improve regression performance. Deviating from the paper and to ensure total separation, we utilized the position -17.992 mm exclusively as a validation subset.

The model was trained for up to 500 epochs using the Adam optimizer with a learning rate of 6×10^{-4} and a weight decay of 9×10^{-5}. We employed SmoothL1Loss as the training criterion to provide a robust gradient response that is less sensitive to outliers than the standard MSE. We implemented a Mixup strategy during training ($\alpha = 2.1$) to combat the discrete nature of the training labels. A ReducedLROnPlateau scheduler was utilized to decay the learing rate when validation loss plateaued.

The final training run demonstrated excellent convergence, as illustrated in the loss and RMSE plots.

- **Loss Convergence:** Both training and validation loss (scaled MSE) showed a consistent downward trajectory with the validation loss curve steadily below the

Model	Data Set	With Amp	With PCA	Extrapolation RMSE ($y = -41.900$ mm)	Overall Test RMSE [mm]
Linear Regression	Augmented	No	No	2.0338	1.8988
Ridge Regression	Reduced	No	No	1.1454	1.6291
Lasso Regression	Full	No	No	0.9377	1.1622
DNN (2-layer)	Reduced	No	No	1.7537	1.5831

Table 1. Extrapolation ($y = -41.900$ mm location) RMSE [mm] for the best configuration of each model based on Overall Test RMSE [mm].

training loss curve. This gap is a direct result of the heavy noise augmentation applied to the training set, making the training task significantly more difficult than the "clean" validation task.

- **RMSE Performance:** The model achieved a remarkably low Validation RMSE of 1.3659 mm and a Training RMSE of 1.7990 mm at convergence. The loss curves described are shown in Figure 2.

Upon evaluating the best-saved model on the HOS dataset, the Two-Branch CNN achieved a HOS RMSE of 1.611 mm. This represents a 7.4% improvement over the benchmark 1.7411 mm. The success of this architecture demonstrates that isolating the pulse shape features from sequenced patterns through a weight-sharing encoder is an effective strategy.

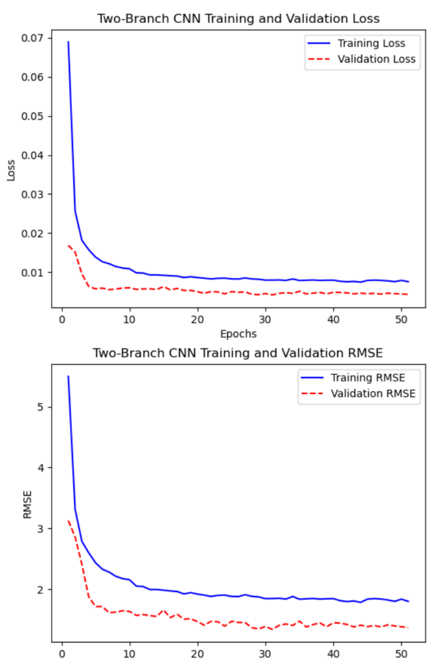

Figure 2. Two-Branch CNN Training Loss Curves. The top plot shows the decrease of L1 loss over training epochs, while the bottom plot shows the corresponding RMSE changes.

4.3. Thermal Attention

Using the 80/20 percent split for the held-out set, we obtained a RMSE of 0.4621 mm. The original HOS achieved an error of 1.7526 mm. The tuned parameters for each HOS are shown in Table 2.

HOS Type	Original FAIR	80/20 Split
Hidden Dimension Size	10	300
Base Learning Rate	5e-5	4e-4
Thermal Learning Rate	2e-3	8e-2
RMSE (mm)	1.7526	0.4621

Table 2. Tuned parameters for the thermal attention model.

The original HOS-trained model's RMSE of 1.7526 mm was a decrease in performance by 0.6%, which measures up to around the same quality off a learned solution given that this distance difference of 11.6 μm is within the standard deviation of the position interactions (of 470 μm) outlined in [1].

The 80/20% split performed better with a RMSE of 0.4621 mm, a 73.45 % reduction in error. It appears that the model was able to perform better with access to a greater variety of location classes, especially since the switch from the first hold out method to the split method increases the number of location classes to 13 from 10.

There is something to be said about the number of hidden dimensions in the best hyperparameters for the original HOS, which was an astonishing 10 compared to 300 for the other data split. This could signify that the more location classes in the input dataset, the harder time the model has generalizing a solution that works for all location classes. This could be evidence of an overfit on the smaller model, as the model may not really be learning to generalize to the held out points but rather report an answer of the nearest point in the training set, especially since the RMSE of the held out points with their nearest neighbors of the training set is approximately 0.678 mm, within the margin of error of the smaller model.

The model trained on the split data had an error below this number, which is good evidence of learning taking place. The learning curves for the thermal attention model are shown in Figure 3. The better performing model has more jagged train and validation curves. This is partially due to the more aggressive learning rates chosen, especially for the thermal parameter learning rates. To properly learn β, each attention head must cover a large feature space, as β can traditionally range from 0.001 to 100. The high learning rate for the thermal parameters was deliberately selected to allow the model to cover enough feature space to find sufficient weightings but lead to a decrease in model stability. There is room for future work to attempt to mitigate this instability, like using a different loss criterion than MSE for this model as it is more sensitive to outlier data.

Adding in the augmented dataset decreased performance in both models, with resulting RMSE values of 4.8650 and 3.4672 mm for each input dataset, respectively.

Overall, this model has its advantages and disadvantages. It did not show much improvement for the original HOS, but did show significant increases in performance with a more representative dataset. This model was a very interesting proof-of-concept to develop and implement, and we see it as a possible starting point for building models tied to physics-related problems, there are some concerns about how deep learning models learn relationships and how learned relationships map to physical systems and mechanisms.

5. Conclusion

In the implementation of the Two-Branch CNN, by decoupling per-detector pulse shapes from cross-detector spatial arrival patterns, the model utilized a weight-sharing convolutional encoder to maximize the information extracted from each interaction event. This architectural design, combined with Mixup training strategy allowed the model to maintain a low parameter count (4,769 parameters) while improving generalization. Ultimately, the Two-Branch CNN achieved a HOS RMSE of 1.611 mm, representing a 7.4% improvement over the previous DNN benchmark of 1.741 mm.

The thermal attention model is a novel approach that utilizes thermal physics to create a model that learns parameters related to the physical processes that govern the data. Additionally, each attention head learns its own governing parameters, allowing the model to change the influence and shape of its internal softmax attention. This model achieved a best RMSE of 0.4621 mm, a 73.45 % reduction in error.

The implemented models we designed and implemented have lower parameter counts compared to some of the deeper models tested in [1], which could have tens of thousands of parameters. This improvement from architecture alone suggests that a model tailored specifically to a unique learning problem such as this can have significant impact on the effectiveness of a machine learning approach and should be considered when attempting to perform such a task.

Our method of data augmentation failed to improve the error of all of our methods, and it was not used to obtain our final values. This was not as expected, but could be due to a number of issues, such as the dataset not having many interaction location classes, or a poor selection of noise intensities used to augment the data.

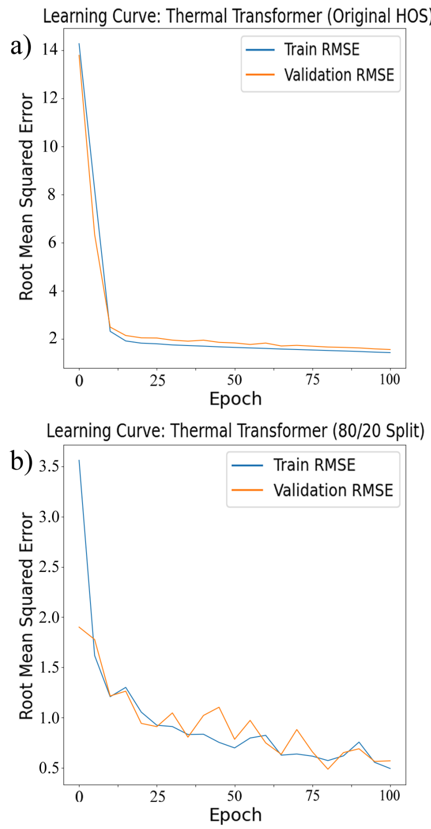

Figure 3. a) The train and validation learning curves for the model trained on the original HOS. b) The train and validation learning curves for the model trained on the original HOS. The aggressive learning rates lead to a more jagged curve as the attention heads all optimize their thermal parameters.

6. Potential Future Work

The method of data augmentation used may have benefit in attempting to train a model to *explicitly* to identify weak interacting massive particles (WIMPs) due to the Am-241 radiation source in [1] emitting particles with more energy than the perturbations a WIMP interacting with a detector would. We believe there may be value investigating how artificially increasing the SNR can help train a model to identify a target particle that would have similar SNR.

We would be enthusiastic to have our methods applied to a more broad and current dataset taken from the same sensor to see if the models we showcase still hold up or even improve with a broader dataset. This could be conducted as a entire separate training from scratch or as a transfer learning problem.

7. Work Division

This project was a collaborative effort involving significant contributions from all team members across all phases of development. While specific technical components were led by individuals as outlined in Table 3, the final integration and paper synthesis were performed collectively.

A supplementary zip file containing the code we have written and utilized is provided to accompany this paper. It contains our additions to the original GitHub repository presented in [1], and we have provided some pre-trained models.

Student Name (GT ID)	Contributed Aspects
Zachary Allen (zallen34)	• Proposed project concept • Systematic framework to asses spatial variation in position reconstruction quality • Added per-class RMSE helper functions to all relevant models • Updated original codebase from Python 3.6 to 3.12 via new conda environment .yml and corrections to outdated pandas functions.
Robert St. Clair (rclair7)	• Unscrambled original data columns with a stochastic optimizer. • Data augmentation via noise addition. Proposed method and developed scripts. • Custom attention model research and the development of attention model architecture. • Full implementation of attention model in code, including the custom thermal self-attention design. • Initial proposal of CNN architecture.
Utsab Mitra (umitra7)	• Full implementation of Two-Branch CNN. • Improved on the proposed CNN architecture to better suit the task. • Mixup batch training strategy.
All	• Paper writing and editing. • Figures within the text. • Exploration of learning strategies and model candidates. • Iterative collaboration.

Table 3. Contributions of team members.

References

[1] Priscilla Cushman, Miriam Colson Fritts, Aidan Chambers, Abhishek Roy, and T. Li. Strategies for machine learning applied to noisy hep datasets: Modular solid state detectors from supercdms. 2024. 1, 2, 5, 6, 7

[2] N. F., Fritscht, , E. R., and Carlson. Monotone piecewise cubic interpolation. *SIAM Journal on Numerical Analysis*, 17:238–246, 1980. 2

[3] Charles Kittel and Herbert Kroemer. *Thermal Physics*. W. H. Freeman and Company, New York, 2 edition, 1980. 3

[4] Yann Lecun, Patrick Haffner, Yoesoep Rachmad, and Leon Bottou. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86:2278 – 2324, 12 1998. 3

[5] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. Imagenet large scale visual recognition challenge, 2015. 3

[6] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition, 2015. 3

[7] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Ł ukasz Kaiser, and Illia Polosukhin. Attention is all you need. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. 3

[8] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric, 2018. 4